value to calculate the AP score in the evaluation phase (Section 3.3).

3 Experimental Settings

In this section, we first provide an overview of the benchmark datasets and the corresponding evaluation settings. Then the baseline models and the implementation details are presented in the following subsections, respectively.

3.1 Datasets

Given the diverse data formats across tasks, we unify the downstream data into two formats:

- *Question-Answering*: Given a question, the model directly predicts its answer. We include **ParaRel** (Elazar et al., 2021), **HotpotQA** (Yang et al., 2018), **SelfAware** (Yin et al., 2023), **HaluEval** (Li et al., 2023a), **FalseQA** (Hu et al., 2023), and **NEC** (Liu et al., 2023) in our experiments.
- *Multiple-Choice*: Given a question with several choices, the model chooses one option. We include **MMLU** (Hendrycks et al., 2021), **WiCE** (Kamoi et al., 2023), and **FEVER** (Thorne et al., 2018) in our experiments.

More information about data processing and evaluation is described in Appendix A.1.

We design two types of experiments:

- *Single-task*: The single-task experiments verify the effectiveness of learning on individual tasks. We conduct experiments on ParaRel and MMLU datasets, respectively. We manually split the datasets into the training set, in-domain test set, and out-of-domain test set. Each dataset contains domain annotations for their questions. Questions in the first half of the domains are selected as in-domain while the remaining are out-of-domain.
- *Multi-task*: The multi-task experiments aim to evaluate the model's generalization performance. We choose five datasets - ParaRel, MMLU, WiCE, HotpotQA, and FEVER, and mix them to construct a new training dataset. As for testing, we evaluate the performance on their corresponding test set (in-domain) and an unseen test set (i.e., HaluEval) (out-of-domain).

3.2 Baselines

We consider three baseline models as follows:

- Pretrain-T: Evaluate the performance of original pre-trained checkpoints on the entire test set.
- Pretrain-W: To verify the effectiveness of willingly answered questions, we evaluate the performance of the original pre-trained checkpoints on the test set that our fine-tuned models are willing to answer. Intuitively, if the willingly answered questions are within the base model's knowledge, this baseline should perform well.
- Vanilla: Fine-tune the model on D with all questions and ground-truth labels. This is the traditional instruction tuning method.

3.3 Evaluation

For models that could only output either the answer or an unknown expression, we evaluate the questions that our model is willing to answer. The accuracy is calculated as follows:

$$\text{accuracy} = \frac{\text{\# of correctly answered questions}}{\text{\# of willingly answered questions}}. \quad (3)$$

For R-Tuning, because it could output both the question's answer and the uncertainty, we first prompt the model to provide an answer and then prompt it to provide its uncertainty. Then we can evaluate the precision-recall tradeoff based on the uncertainty and prediction performance. We introduce the Average Precision (AP) score, which measures the precision in identifying and ranking relevant predictions. AP score originates from the object detection field (Everingham et al., 2010) by ranking the prediction results by confidence from high to low and calculating the precision at each threshold. The AP score is the average of these precision scores, which is calculated as follows:

$$AP = \sum_{k=0}^{n-1} (R(k+1) - R(k)) \times P(k), \quad (4)$$

where n is the number of data, k is the number of data we select for the current threshold. P and R denote precision and recall, which are defined as

$$P(k) = \frac{\text{\# of correct answers above k-threshold}}{\text{\# of answers above k-threshold}}, \quad (5)$$

$$R(k) = \frac{\text{\# of correct answers above k-threshold}}{\text{\# of correct answers}}. \quad (6)$$

An ideal model predicts the correct answers with high confidence and the hallucinated wrong answers with relatively low confidence, leading to a high AP score. On the other hand, the AP score is low if the model predicts every answer with high confidence, as the precision at every threshold will not be high and the average will be relatively low.

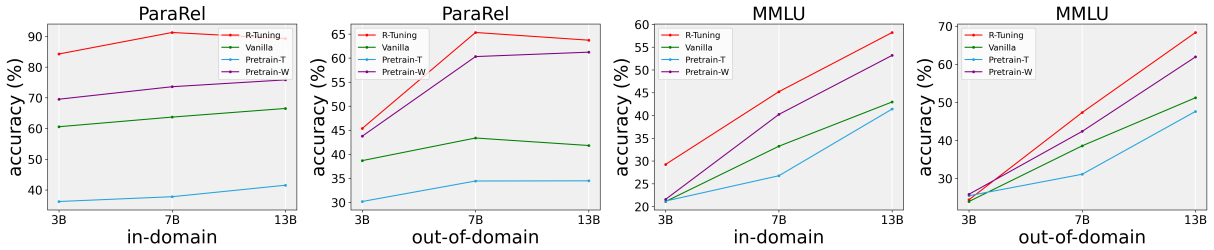


Figure 3: Single-task experiments on ParaRel and MMLU datasets with accuracy (%). R-Tuning is calculated on the willingly answered questions. Pretrain-W is verified on these questions. Others are calculated over the entire dataset.

Dataset	Domain	Models	R-Tuning	Vanilla
ParaRel	ID	OpenLLaMA-3B	**93.23**	92.89
		LLaMA-7B	**93.64**	93.32
		LLaMA-13B	**94.44**	94.00
	OOD	OpenLLaMA-3B	**69.41**	68.42
		LLaMA-7B	74.61	**78.08**
		LLaMA-13B	**77.30**	64.12
MMLU	ID	OpenLLaMA-3B	**24.96**	24.19
		LLaMA-7B	**59.05**	58.16
		LLaMA-13B	**68.87**	51.93
	OOD	OpenLLaMA-3B	24.75	**26.08**
		LLaMA-7B	**68.69**	66.38
		LLaMA-13B	**77.41**	67.38

Table 1: Single-task experiments of R-Tuning and Vanilla on ParaRel and MMLU datasets with AP scores (%). ID and OOD denote in-domain and out-of-domain settings, respectively.

3.4 Implementation

We choose OpenLLaMA-3B (Geng and Liu, 2023), LLaMA-7B, and LLaMA-13B (Touvron et al., 2023) as the base models in our experiments. We use LMFlow[2] (Diao et al., 2023a) to conduct instruction tuning, setting epoch to 1, learning rate to $2e^{-5}$, and batch size to 4. All the experiments are implemented on Nvidia A100-40GB GPUs.

4 Experimental Results

In the main experiments, we conduct single-task experiments to verify the model's refusal-aware answering ability and multi-task experiments to investigate the generalization of refusal ability.

4.1 Single-task Experiments

We first conduct single-task experiments on ParaRel and MMLU datasets. The results are shown in Figure 3 and Table 1. Firstly, we observe that R-Tuning significantly outperforms other baselines by a large margin in terms of accuracy on the questions it is willing to answer, compared with others that simply answer all the questions. The results first demonstrate the effectiveness of the refusal-aware answering ability. We also conclude that R-Tuning answers more questions within its parametric knowledge during pre-training, which is reflected by the high accuracy of Pretrain-W (pretrained model evaluated on R-Tuning's willingly answered questions). Overall, it is observed from Table 1 that R-Tuning outperforms Vanilla in terms of the AP score, demonstrating the benefits of only answering the questions that align with the model's parametric knowledge with high confidence. In addition, we find that larger models achieve more improvement compared with baselines as the gap of the AP score becomes larger, indicating good scalability of R-Tuning. The AP score of R-Tuning grows steadily when the model size becomes larger, while the AP score of Vanilla drops in ParaRel (OOD) and MMLU (ID). This comparison shows that Vanilla may suffer from confidence miscalibration problems while R-Tuning is more well-calibrated in terms of confidence. By combining the prediction confidence and certainty confidence to evaluate the output, R-Tuning is more reliable when making predictions.

4.2 Multi-task Experiments

The results of multi-task experiments are shown in Figure 4. Overall, R-Tuning consistently outperforms all baseline models in terms of the AP score on both ID and OOD tasks, demonstrating its superiority by introducing the refusal-aware dataset. A higher AP score signifies that the R-Tuning has successfully ranked correct answers higher than incorrect answers, demonstrating its effectiveness in accurately identifying the desired predictions. Especially, on the unseen dataset HaluEval-QA, R-Tuning also achieves a higher AP score and demonstrates its ability to express certainty to questions from other distributions, and such ability can be generalized well. The experiments on multi-task

[2] https://github.com/OptimalScale/LMFlow

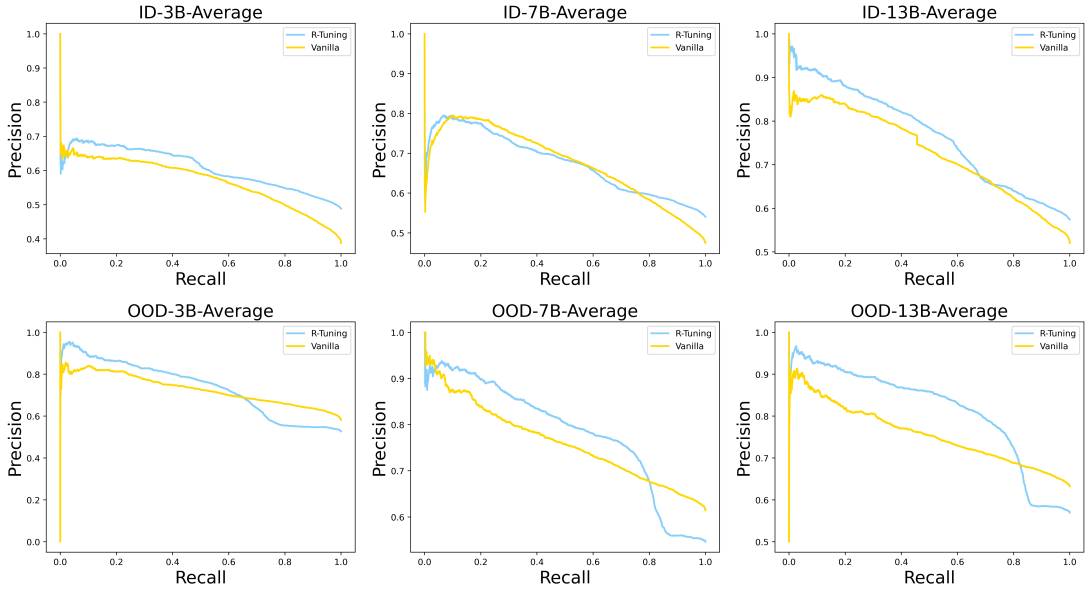


Figure 4: Multi-task experiments on the average of five in-domain (ID) datasets (ParaRel, MMLU, WiCE, HotpotQA, and FEVER) and one out-of-domain (OOD) dataset (HaluEval-QA) with the AP curves.

datasets tell us that the refusal is a kind of meta-skill of models and could be enhanced by several different datasets. We provide the detailed AP scores and curves for different datasets and model sizes in Table 13 and Figure 8 in Appendix A.10.

In summary, R-Tuning reduces hallucinations by disregarding inquiries outside of the model's knowledge domain. Meanwhile, R-Tuning performs well with inquiries that are aligned with the model's parameterized knowledge. The better AP score demonstrates a good trade-off between precision and recall and the performance on multi-task experiments demonstrates the generalization potential of refusal-aware answering ability.

5 Analysis

In this section, we first introduce a variant, R-Tuning-U, which adopts an unsupervised identification strategy for R-Tuning. Then we provide an interpretation from the uncertainty perspective for R-Tuning. In addition, we verify the refusal ability on unanswerable questions, which should not receive answers from the model. More case studies are shown in Table 9 in the Appendix for qualitative analysis. Further analysis of the perplexity (Section A.6) and uncertainty of the training datasets (Section A.7) demonstrates the effectiveness of our proposed method.

5.1 Unsupervised Identification

During the refusal-aware data identification process, we apply a supervised way to identify unknown questions by comparing the predictions and labels. In this section, we introduce an unsupervised identification method, R-Tuning-U, where the refused questions are determined by the uncertainty of the model. Specifically, R-Tuning-U queries the model M k times and calculates the uncertainty u across k predictions, which is calculated by the entropy based on k answers as follows:

$$u = -\sum_{j=1}^{k} p(a_j|q) \ln p(a_j|q), \tag{7}$$

where $p(a_j|q)$ is the frequency of a certain predicted answer a_j given a question q.

Then the questions could be ranked according to the uncertainty score u. For the 50% most uncertain questions, we append the ground truth label and uncertain expression (i.e., uncertain set D_0), while the remaining (i.e., certain set D_1) are appended with the ground truth answers with certain expressions. We set the temperature to 0.7 and $k = 10$ in our experiments. We compare the performance with the R-Tuning on the ParaRel and MMLU datasets, and the results are shown in Table 2. It is observed that R-Tuning-U generally achieves a higher AP score, which reveals the feasibility of constructing refusal-aware training data by uncertainty. Comparing the output of the pre-trained model with the ground-truth answer is not the only way to evaluate its parametric knowledge. Uncertainty can also be an indicator of whether the pre-trained model is familiar with the knowledge.